

Evolution Lab

GENE 4230L

Dr. Kelly Dyer
Dr. John Wares

Fall 2009

Course Information

Prerequisites: You should have already taken GENE 3000 (Intro to Evolution); may be taken as corequisite with instructor permission.

Instructors: Dr. Kelly Dyer, kdyer@uga.edu; Dr. John Wares, jpwares@uga.edu; office hours by appointment only. Teaching assistant, Monica Poelchau, poelchau@uga.edu; Lab manager, Carmen Rodriguez, crodrig@uga.edu

Course Description: Introductory lab to explore the mechanisms of evolution – mutation, selection, drift, gene flow, behavior. You will need to have a firm grasp of Mendelian genetics and the basics of macromolecular structure and biochemistry involved in genetics. Basic algebra and statistics will be required. Few of the topics can be covered in great detail, but this course will expose you to a variety of methods used in modern evolutionary genetics.

Grading: your overall grade in this class will be based on the following distribution:

Attendance and participation, **5 points per class**. Unexcused absence, late arrival, disruptive behavior will be counted against you.

Quizzes, **10 points**, to be completed prior to most classes. Material will be based on reading necessary to participate in lab that day.

Midterm exam, **100 points**

Reports, **100 points each** (4 over entire semester)

Final exam, **100 points**

The final number of points may vary. Late assignments will be docked 10% per day. A "B" is considered the average grade in this class, with the approximate grade distribution following the quartiles: A is 92.5 or above, A- is 90-92.49, B+ is 87.5-89.99, B is 82.5-87.49, B- is 80-82.49, and so on. If necessary we will curve to class advantage. No makeup exams or quizzes will be given; if you have legitimate excused absences with documentation, your grade will be based on remaining scores.

Textbook: Lab handouts and papers for discussion will be distributed via WebCT. There is no required textbook for this course. However, we suggest you keep an evolutionary biology textbook handy as a reference. In addition, we have placed the following book at the bookstore, which you may find it very useful as a resource for analyses, lab write-ups, and presentations.

A Student Handbook for Writing in Biology, 3rd Edition. 2009. By Karin Knisely.

Academic Honesty: All students are responsible for maintaining the highest standards of honesty and integrity in every phase of their academic career. The penalties for academic dishonesty are severe *and ignorance is not an acceptable defense*. All academic work must meet the standards contained in "A Culture of Honesty". Students are responsible for informing themselves about those standards before performing any academic work. The link to more detailed information about academic honesty can be found at www.uga.edu/ovpi/honesty/acadhon.htm. You will be working closely with partners in this lab but your work must be your own, reflect your own thought, show your own math and reasoning, and so forth. We will compare your results to ensure you collaborate appropriately. Please note that we have caught plagiarists in previous years of this class and have been strict but maintained it as a private matter. If it happens again we will not be so lenient. Copying text and ideas from other sources without proper credit is illegitimate and will be punished. You have been warned.

The syllabus is a general plan for the course; deviations announced to the class by the instructors may be necessary.

Syllabus

Lab	Date	Topic	
1	Mon Aug. 17	Introduction; Lab Skills and Dilution Study	
2	Wed Aug. 19	Luria-Delbrück I	
3	Mon Aug. 24	Luria-Delbrück II	
4	Wed Aug. 26	Identifying Mutations I	
5	Mon Aug. 31	Identifying Mutations II	(L-D write up due)
6	Wed Sep. 2	Probability and Hardy-Weinberg	
	Mon Sep. 7	Labor Day – no class	
7	Wed Sep. 9	Identifying Mutations III	
8	Mon Sep. 14	Fitness of Mutations I	
9	Wed Sep. 16	Fitness of Mutations II	
10	Mon Sep. 21	Fitness of Mutations III	
11	Wed Sep. 23	Fitness of Mutations IV	
12	Mon Sep. 28	Fitness of Mutations V	
13	Wed Sep. 30	Population Diversity I	
14	Mon Oct. 5	Population Diversity II	(Fitness write up due)
15	Wed Oct. 7	Population Diversity III	
16	Mon Oct. 12	Population Diversity IV	
	Wed Oct. 14	MIDTERM	
17	Mon Oct. 19	Quantitative traits	
18	Wed Oct. 21	Reproductive Isolation I	
19	Mon Oct. 26	Reproductive Isolation II	
20	Wed Oct. 28	Meta-diversity I	
21	Mon Nov. 2	Meta-diversity II	
22	Wed Nov. 4	Meta-diversity III	
23	Mon Nov. 9	Phylogeny I	
24	Wed Nov. 11	Reproductive Isolation III	
25	Mon Nov. 16	Phylogeny II	(RI write up due)
26	Wed Nov. 18	Phylogeny III	
	Nov. 23,25	Thanksgiving – no class	
27	Mon Nov. 30	Meta-diversity IV	
28	Wed Dec. 2	Meta-diversity V	
29	Mon Dec. 7	Meta-diversity VI	
	Wed Dec. 14	FINAL EXAM 12 PM	(Meta-diversity write up due)